

## Exp No: 17

### UART INTERFACING USING INTERRUPT IN KEIL

#### OBJECTIVES

- To understand UART serial communication in the AT89C51 microcontroller.
- To configure and enable serial interrupts for receiving data through UART.
- To display the ASCII value of the received character on Port 1.

#### PROGRAM

```
#include <reg51.h>

void serial_ISR(void) interrupt 4
{
    if(RI)
    {
        P1 = SBUF;
        RI = 0;
    }
}

void main()
{
    TMOD=0x20;
    TH1=0xFD;
    SCON=0x50;
    TR1=1;
    ES=1;
    EA=1;
    while(1);
}
```

# OUTPUT

The screenshot displays the uVision IDE interface for a project named "Experiment 17\UART\_INTERRUPT.uvproj". The main window shows the disassembly of the program, with the following assembly code visible:

```
18: while(1):  
C:0x080F 80FE Sjmp C:080F  
133: MOV R0,#IDATALEN - 1  
S:0x0811 787F MOV R0,#0x7F  
134: CLR A
```

The source code window shows the following C code:

```
1 #include <reg51.h>  
2 void serial_ISR(void) interrupt 4  
3 {  
4     if(RI)  
5     {  
6         P1 = SBUF;  
7         RI = 0;  
8     }  
9 }  
10 void main()  
11 {  
12     TMOD=0x20;  
13     TH1=0x7D;  
14     SCON=0x50;  
15     TR1=1;  
16     ES=1;  
17     EA=1;  
18     while(1);  
19 }
```

Several configuration windows are open:

- Parallel Port 1:** Port 1 is set to 7 Bits, 0 Parity, and Pins are 0x41.
- Interrupt System:** A table showing interrupt sources and their status.
- Serial Channel:** Mode is 8-Bit var Baudrate, SCON is 0x54, SBUF is 0x00, SM2 is checked, REN is checked, TB8 is checked, RB8 is checked, Baudrate is 20833, and TI and RI are unchecked.

The Command window shows the following output:

```
Running with Code Size Limit: 2K  
Load "C:\Users\M.POOJITH GANESH\OneDrive\embedded\Experiment 17\Objects\UART_INTERRUPT
```

The bottom status bar shows the following information:

ASH ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE